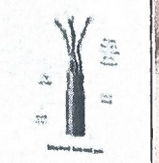
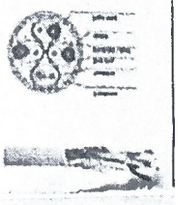
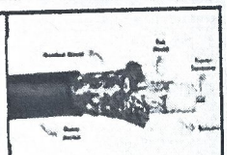
Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use	Image
UTP	Category 3	10 bps	**Advantages** • Cheaper in cost • Easy to install as they have a smaller overall diameter. **Disadvantages** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet	
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category6,6a	10Gbps	**Advantages** • Shielded. • Faster than UTP. • Less susceptible to noise and interference **Disadvantages** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	
Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile • **Disadvantages** • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections	
fibre optics cable	Single mode Multi mode	100Gbps	**Advantages** • High speed • High bandwidth • High security • Long distance **Disadvantages** • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters	

Take a print out the diagram below or have it handy as a reference

Straight through network cable: both sides should be A

Exp-2

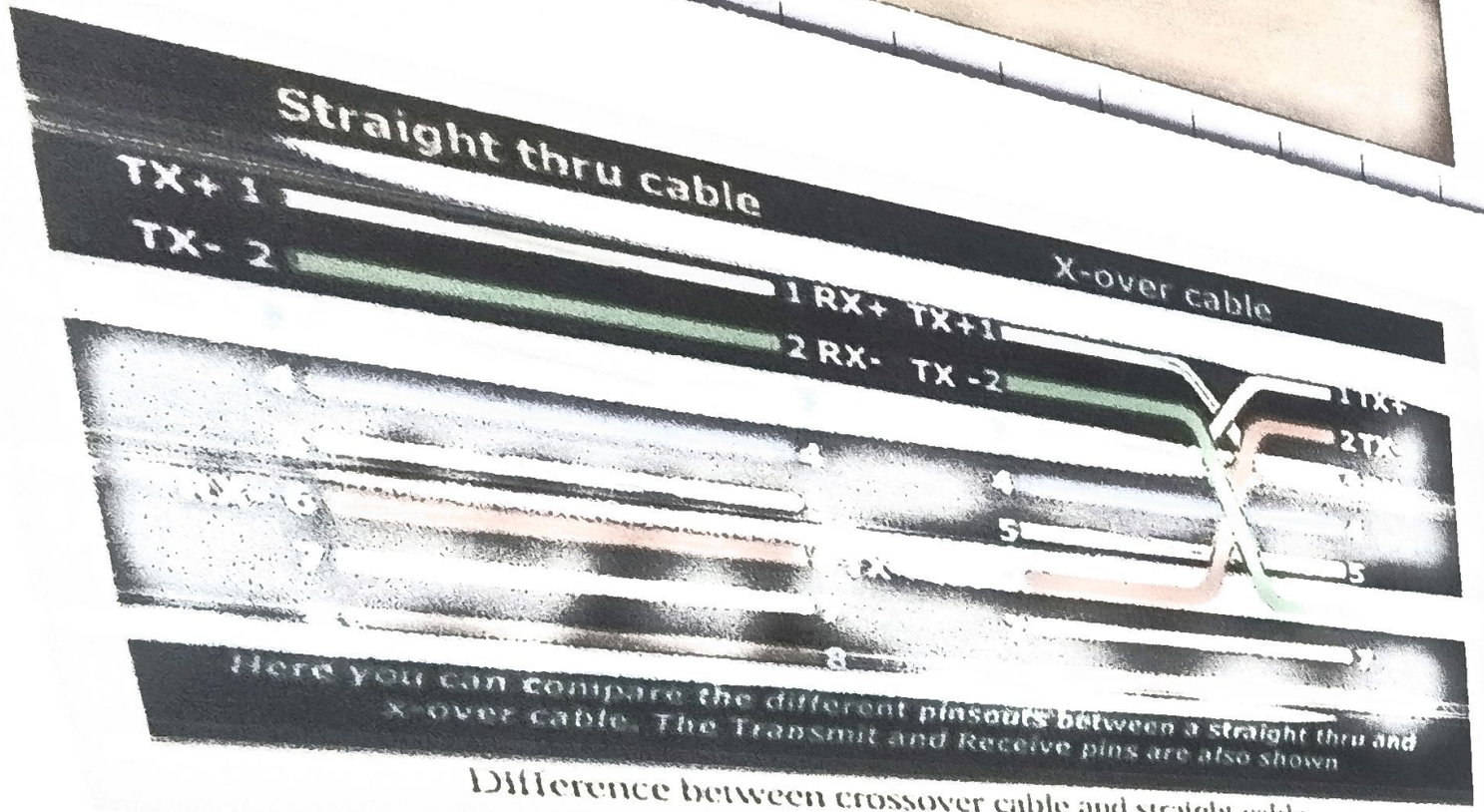
Aim: Study of different types of Network cabeels.

a) understand different types of Network cable.
Different types cable used in networking.
1) unshielded twisted pair (UTP cable)
2) Shielded twisted pair (STP) cable
3) Carnvial cable.
4) Fibre optic cable.

b) Make you own Ethernet (rose-over cable/ Straight cable.

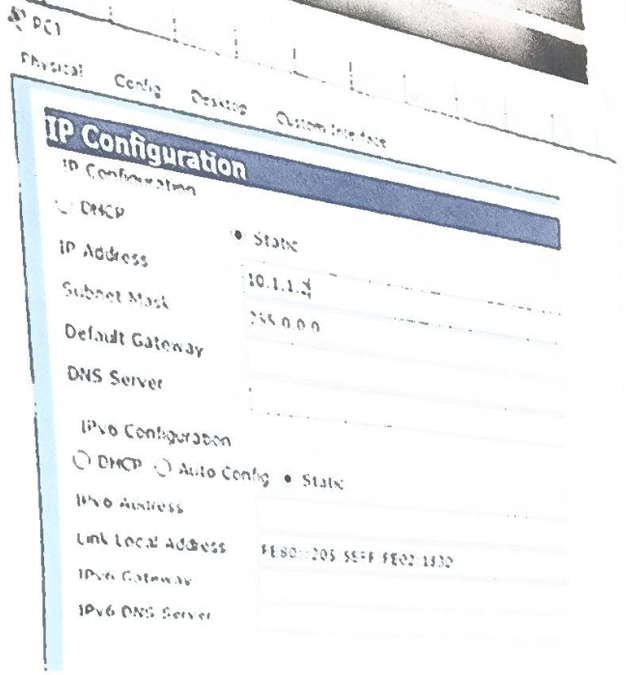
Tools and Parts needed.
+ Ethernet cable is certified for gigabit support but 5 cabling works as well just over short distance.
+ Two RJ45 Plugs.
+ Optional two Plugs sheilds.
+ A crimpling tool. this is all in one Networking tool shaped to push down the pins in the plug and strip and cut the sheilding of the cables

Step1: To start construction of device begin by threading sheilds onto the cabe.
Step2: Next, strip aprox 1.5 cm of cable sheilding from both ends. the crimping tool has a round area to complete task.



Difference between crossover cable and straight cable





PC0

Step 3:

After you will need to untangle the wires there should be four twisted pairs Referring back to the sheet arrange them from top to bottom. one end should be in arrangement A and other in B.

Step 4:

Once the order is correct but them together is in line and if there are any that stick out farther than others skip them create and even level.

Step 5:

Next push the cable right. the notch at the end of the plug needs to just over the cable sheilding and if it isn't that means that you stripped off too much sheilding simple skip the cable back a little mor

Step 6:

After the wires are securly inside the plug insert into the crimping tool and push down.

Step 7: Lastly repeat for the other end using diagram B (to make a crossover cable) using diagram A (to make a straight through cable)

Result: Thus the study of different type of cable is done successfully